

Sully Palaia

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github.com/sullypalaia

Experience

**Robotics Intern | Colorado School of Mines
ARIA Lab**

Summer 2025

- Worked with and developed software for the autonomous navigation of Boston Dynamics robot SPOT using ROS2 and RTAB-Map
 - Collaborated with a software team and coordinated development with hardware teams and other software teams
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Education

California Polytechnic State University, San Luis Obispo

B.S. Computer Engineering | Expected
Graduation: Spring 2030

Evergreen High School | Evergreen, CO

August 2022 – May 2026

Projects

**UAV-assisted Trail Route Design Automation
Tool/Simulation**

Dec. 2025 - Feb. 2026

- Created a Gazebo simulation environment involving the autonomous (or optionally manual) movement of a UAV with an attached LiDAR sensor.
- Built a tool to automatically create a trail route based on a least-cost path architecture implemented with Dijkstra's algorithm and Point Cloud Library
- Designed a landmark-based system in which a UAV will intentionally avoid areas designated as obstacles, such as active bird nests, when collecting data and generating a route

Game Engine

July 2026 – present (WIP)

- Built a tool that automatically generates GLSL shaders from a set of components (position, color, transforms, etc...)
 - Using data-oriented design principles to implement an entity component system using EnTT
 - Creating a renderer using OpenGL 4.5 with support for instanced rendering, broader multi-draw, and other optimizations
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Skills

- C++ (20)
- Python
- ROS2
- Lua
- OpenGL (4.5)